

A Pragmatic Approach to the Analysis of DNA Histograms With a Definable G1 Peak

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A method for DNA histogram analysis is described that depends only on the simple assumption that the data are normally distributed and a requirement that a G1 peak is present. A probability density function was derived from the assumption that extracted the S-phase component from the whole histogram. The model was tested with simulated data, and good agreement between predicted and known proportions in G1, S, and G1+M was found. Good agreement was also found between duplicates of experimentally derived data. Some systematic errors are present in the analysis of certain types of histograms. However, these result in small errors when

compared with biological and experimental variation and are less than the average of algorithms in current use.

The program required only two queued requests, those of the start and the end channels over which the analysis is to be performed. The algorithms perform rapidly on a microcomputer with only 28K addressable memory. Only two failures occurred in over 350 analyses and the method can be used for drug- and radiation-perturbed populations as well as with unperturbed.

Key terms: Flow cytometry, DNA histogram analysis

A number of models have been proposed for the analysis of flow cytometric DNA histogram data (1, 2, 4-9, 11-18, 25, 28-29) and a comparative review of the various methods has been published (3). As expected, some models performed better than others, not only with simulated but also with experimentally derived data; however, none was ideal. The models that performed best tended to be those requiring a large mainframe computer, as they generally contained a large number of variables for which a solution had to be found. These include the position of the mean of the G1 and G2+M peaks, the standard deviations of these peaks, the age distribution of the population (22), the rate of DNA synthesis, and the relative durations of the G1, S, and G2+M phase times with their standard deviations. Thus far, 12 variables have been defined, all of which may have to be considered simultaneously, which is not a trivial task, even with a large computer. Herein lies one of the central problems. The experimental DNA histogram contains, at best, two peaks and a trough corresponding to G1, G2+M, and S-phase, respectively. This is a very simple data set with which to compute 12 variables, and this must be totally inadequate compared with the complexity of the biology.

The work presented in this paper was undertaken to simplify the analysis of DNA histograms and to produce

a robust method that gives results rapidly (within 30 s) on a microcomputer with only 28K addressable memory.

THEORY Assumptions

In our attempt to produce a "minimum assumption and computing" model (MAC) we have only assumed that the data are normally distributed.

The Computer Program

The method of analysis will be illustrated by a description of the program flow diagram and the components of its various steps.

Step 1. Eliminate high-frequency noise, if this exists, by "spreading" the data with a constant standard deviation of 0.75 channels and performing the appropriate summation (21).

Step 2. Input the start and end channels over which the analysis is to take place.

Step 3. Compute a first approximation for the mean and standard deviation (SD) of the G1 peak. These are achieved by finding the channel containing the maxi-

imum frequency to give the mean and by finding the width of the distribution at 60% of the maximum height to the left of the mean to give the SD.

Step 4. Use these initial starting values to perform an iterative least-squares fit between the normal curve and the data over a range of -3 to $+1$ standard deviations about the mean. This range was selected to minimize the effect of any "contamination" due to the overlapping S-phase distribution to the right of the mean.

Step 5. Compute the mean and SD of the G2+M peak. The program finds the highest point in the histogram data array starting at $1.75 \times$ G1 mean, and this is taken as the first approximation for the G2+M mean value. The SD is then calculated as the width of the distribution at 60% of the maximum height to the right of the mean.

Step 6. These initial values are then used for the iterative process (identical to step 4) to calculate a least-squares best fit between the Gaussian curve and the experimental data over a range of -1 to $+3$ SDs. This minimizes the contribution of the overlapping S-phase component to the left of the G2+M.

Step 7. Construct a probability distribution for S-Phase, P_s , such that when the frequency in channel x of the data set is multiplied by its corresponding value of $P_s(x)$ we obtain the number of S-phase cells in that channel. The form of this distribution is given by:

$$P_s(x) = \int_{-\infty}^x \text{ERF}(G1(x) - kG1) - \int_{-\infty}^x \text{ERF}(G2(x) + kG2) \quad (1)$$

where $\text{ERF}(Z)$ is a numerical integration routine for the error function that computes the area under the normal curve for $-\infty$ to Z , where $Z = (x - \bar{x})/\text{SD}$ (10). In the equation shown above,

$$G1(x) = [\text{channel}(x) - G1 \text{ mean}]/\sigma_{G1} \\ \text{and } G2(x) = [\text{channel}(x) - G2 \text{ mean}]/\sigma_{G2}$$

where σ_{G1} and σ_{G2} are the standard deviations of the G1 and G2+M distributions, respectively. The constants $kG1$ and $kG2$ determine the positions of the G1/S and SG2+M interfaces within the histogram. In a perfect data set with no dispersion, the probability of finding an S-phase cell in the single G1 channel would be zero and the probability of finding an S-phase cell in channel $G1+1$ would be unity. Thus, the probability of finding an S-phase cell at the G1/S boundary would be 0.5. The constants $kG1$ and $kG2$ position the S-phase probability function within a distributed data set so that the chance of finding an S-phase cell at the G1 and G2+M means will be 50%. This is depicted graphically in Figure 1, where, for simplicity, only the G1 peak and the first part

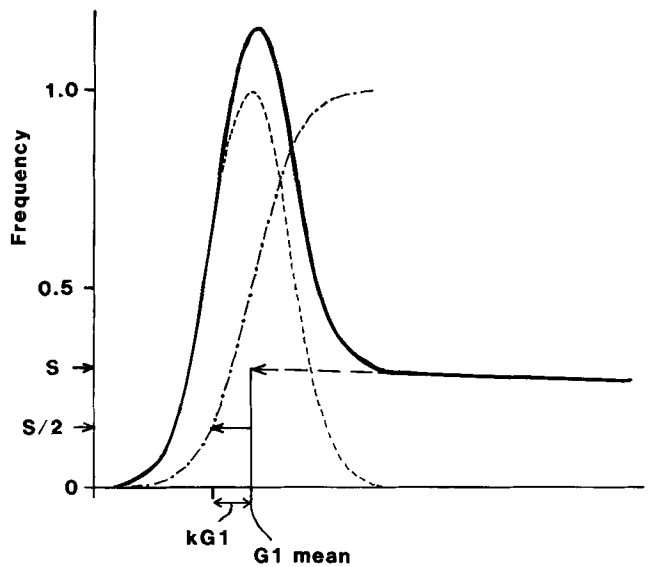


FIG. 1. G1/S region of DNA histogram where the maximum frequency of the G1 component is scaled to unity. The thick, uninterrupted curve bounds the whole data set. The short-dashed curve, mainly to the right of the G1 mean, is the G1 component (Gaussian distributed) and the dot-dashed curve is the cumulative frequency of G1, also scaled to unity. The long-dashed line with negative slope extrapolates the S-phase envelope above $G1 \text{ mean} + 3 \cdot \text{SD}$ to the G1 mean channel and cuts the latter at frequency S . The constant $kG1$ is the number of G1 standard deviation units measured from the G1 mean channel associated with a cumulative frequency of $S/2$.

of S-phase are shown. The thick curve bounds the whole data set and the short-dashed curve to the right of the G1 mean represents the G1 component after subtraction of the S distribution. The dot-dashed curve is the cumulative frequency of the G1 distribution, which is arbitrarily scaled to the unit peak height of the G1 component. This cumulative frequency curve is calculated from the standard deviation of the G1 peak found at step 4 and represents a probability boundary between the G1 and S components with a value of 0.5 occurring at the G1 mean. At $3 \cdot \sigma_{G1}$ above the mean, where the cumulative G1 frequency is unity, the respective probabilities of finding a G1 and S-phase cell are zero and unity. Thus, if we know the frequency of the S distribution at the G1 mean, we can calculate the probability within the whole distribution of finding an S-phase cell at the G1 mean from the cumulative frequency distribution of the G1 compartment. An approximation for this S-frequency can be obtained by extrapolating the envelope of the S-phase above $G1 \text{ mean} + (3 \cdot \text{SD } G1)$ to the G1 mean using regression analysis. This is depicted in Figure 1 as the long-dashed straight line with negative slope. We can now find the point on the cumulative frequency curve associated with a value of $S/2$, (see diagram) and the number of standard deviation units, $kG1$, associated with this point. The value of $kG1$ is found by an iterative process from the following relationship:

$$\text{ERF}(kG1) = S/(G1 \cdot 2) \quad (2)$$

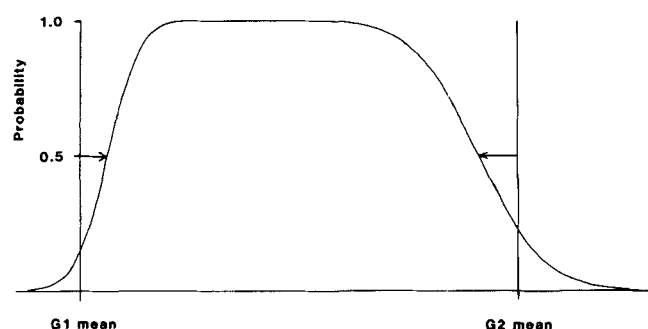


FIG. 2. S-phase probability distribution. When each channel of this distribution is multiplied by the frequency in the corresponding channel of the experimental data set, the S-phase distribution is generated.

where S and G1 are the frequencies of the S and G1 distributions, respectively, at the G1 mean. $kG1$ is set at -3 initially and is incremented by 0.05 sequentially until $ERF(kG1)$ is equal to, or just greater, than $S/(G1 \cdot 2)$. This value of $kG1$ is then taken as the correction factor in equation 1. A similar process is used to find $kG2$.

The form of the S-phase probability distribution is shown in Figure 2, where the shifts of the "leading" and "trailing" shoulders from the G1 and G2+M mean values, due to the correction factors $kG1$ and $kG2$ respectively, are apparent.

When the distribution in Figure 2 is multiplied on a channel-by-channel basis with the corresponding frequencies in each channel of the DNA histogram, we obtain the S component of the histogram. The distribution so generated is such that the frequency of S-phase cells at the G1 and G2+M mean channels is 50% of the frequency of the envelopes extrapolated to the G1 and G2+M mean channels.

Step 8. Subtract the S-phase distribution from the total histogram to give the "uncontaminated" G1 and G2+M distributions.

Step 9. Compute the true mean (M) and variance (V) of these distributions according to the following equations:

$$M = m + \sum \{t(n) \cdot f(n)\} / \sum f(n), \quad (3)$$

$$V = \sum [t(n)^2 \cdot f(n)] / \sum f(n) - [\sum \{t(n) \cdot f(n)\} / \sum f(n)]^2. \quad (4)$$

The term m in equation 3 is the estimate of the means found at steps 4 and 6, $t(n)$ is equal to $n - m$ where n varies from channel i through j and $f(n)$ is the frequency in channel n . Equations 3 and 4 are standard methods of computing the mean and variance of a distribution (19).

Step 10. Compare these new values for the means and SDs of the two distributions with those found at steps 3 and 5. If there is a discrepancy of more than 2.5% between any two pairs of values, return to step 7 with the values found at this step.

When convergence has been obtained, the S-phase distribution is calculated and subtracted from the experimental data set to give the discrete G1 and G2+M distributions from which the proportions in each phase and the G2+M:G1 ratio are calculated.

MATERIALS AND METHODS

Simulated Data

In the report by Baisch et al. (3), a number of synthetic data sets with known proportions in G1, S, and G2+M were analysed blind by all groups involved in the study. A selection of eight of these data sets, where there was a well-defined G1 peak but with widely differing proportions in the cell cycle phases, was reanalysed using the method described here.

Experimental Data

Cell lines. Five different human cell lines were used: HT29 (colon adenocarcinoma), MRC5 (fibroblast line; normal donor), AT5BI (fibroblast line from a patient with ataxia telangiectasia), and SV40 transformed variants (MRC5CVI and AT5BIVA) of the latter cell types. All lines were maintained in monolayer culture and manipulated using standard procedures.

DNA staining. Single cell suspensions were prepared by treating monolayers with versene, or versene-trypsin, and resuspending at 5×10 cells/ml in growth medium. Staining was then carried out with a Triton X-100/ethidium bromide technique (23).

Flow cytometry. DNA determinations were carried out on the Cambridge MRC dual laser flow cytometer (26,27). In most data sets, only two out of seven photodetectors were used to quantitate forward light scatter and red fluorescence (>630 nm), but in some, the 90° light scatter photodetector was also used. The 164-05 argon laser (Spectra-Physics, Mountain View, CA) was tuned to the 488 nm line at 200 mW. After alignment of the instrument with microbeads (Particle Technology Inc.), the data were collected list-mode on an RP07 450 MByte disc via PDP LSI 11-23 and 11-40 computers (Digital Equipment Corporation, Maynard, MA).

Data retrieval and analysis. Following collection, the data were called from disc and analysed by the PDP LSI 11/23 dedicated microcomputer with 28K addressable memory.

RESULTS

Simulated Data

A selection of eight histograms is shown in Figure 3. These were chosen for illustration, as they exhibit widely different forms and fairly marked differences in the proportions in the cell cycle phases. Figure 4 shows the model-predicted proportions in each phase plotted against the known proportions from the analyses in Figure 3. The solid and open circles represent G1 and S, respectively, and the squares represent G2+M. The line has been drawn with unit slope. $\Sigma \chi^2$ for the 24 comparisons was 5.19 with 15 degrees of freedom, $p < 0.0001$ that the deviations of the predicted from the true could have

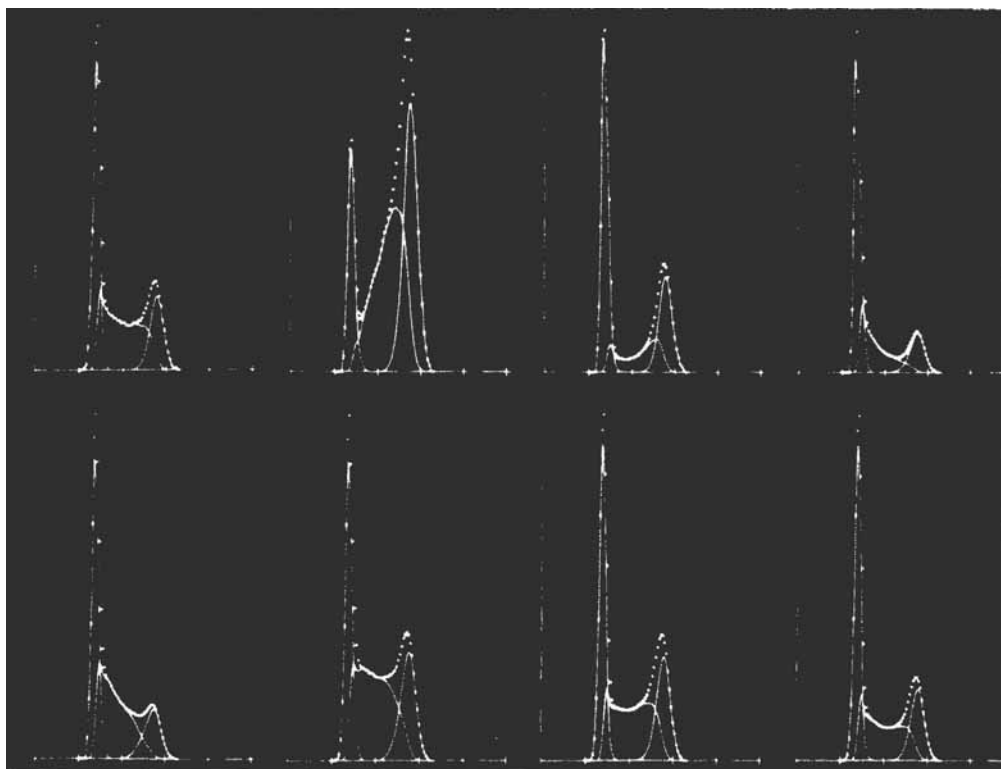


FIG. 3. Illustration of histogram analysis (simulated data). The dots bound the whole of each data set and the curves bound G1, S, and G2+M. Each abscissa division represents 100 channels, and the data

sets were scaled individually to the maximum height of each histogram. For display purposes, these data have been reduced from 10-bit to 8-bit resolution.

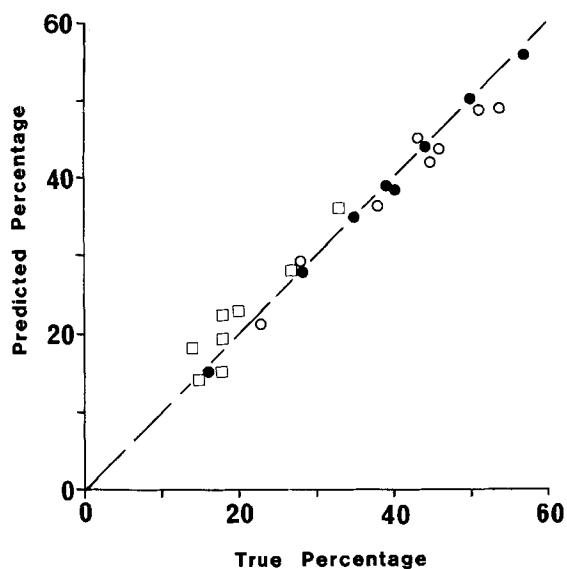


FIG. 4. Model-predicted percentages plotted against known percentages for the data analysed in Figure 3. ●, G1; ○, S-phase, □, G2+M.

arisen by chance. Although 24 points were included in the $\Sigma\chi^2$ calculation, there are only 15 degrees of freedom, as only two values from each histogram can be assigned arbitrarily. The third is fixed by the other two.

The average of the predicted G1 mean values was 138.2 compared with a true value of 138, an error of less

than 0.2%. The G2+M:G1 ratio was 1.993, an error of less than 0.36%. The mean coefficients of variation of the G1 and G2+M peaks were 5.68% and 6.8% respectively, compared with a true value of 5.75% for both.

EXPERIMENTAL DATA

G2+M:G1 ratio. In a series of over 350 DNA histograms from five different cell lines in both control and drug- or radiation-perturbed populations, the G2+M:G1 ratio was 2.009 with 95% confidence limits of 0.015. This overall mean value does not differ significantly from the expected ratio of 2.0, $p > 0.05$. There was somewhat greater variability in the subgroup of drug- and radiation-perturbed data, where the mean was 2.014 with 95% confidence limits of 0.021. The computed proportions in G1, S, and G2+M varied within the ranges of 10 to 90%, 7 to 90% and 5 to 80%, respectively, in a multiplicity of combinations. A selection of six histogram analyses is shown in Figure 5 to illustrate the wide variety of forms that can be analysed by the model.

Duplicate samples. In a number of experiments, duplicate flasks were set up by one of the authors (P.J.S.). These were then stained and run on the flow cytometer by S.H.C. and analysed by J.V.W. The latter two authors did not know which were duplicates. Figure 6 shows the proportions in G1, S, and G2+M from the first sample plotted against the comparable values from the second sample. The closed and open circles depict G1 and S, respectively, and the squares represent G2+M. A total

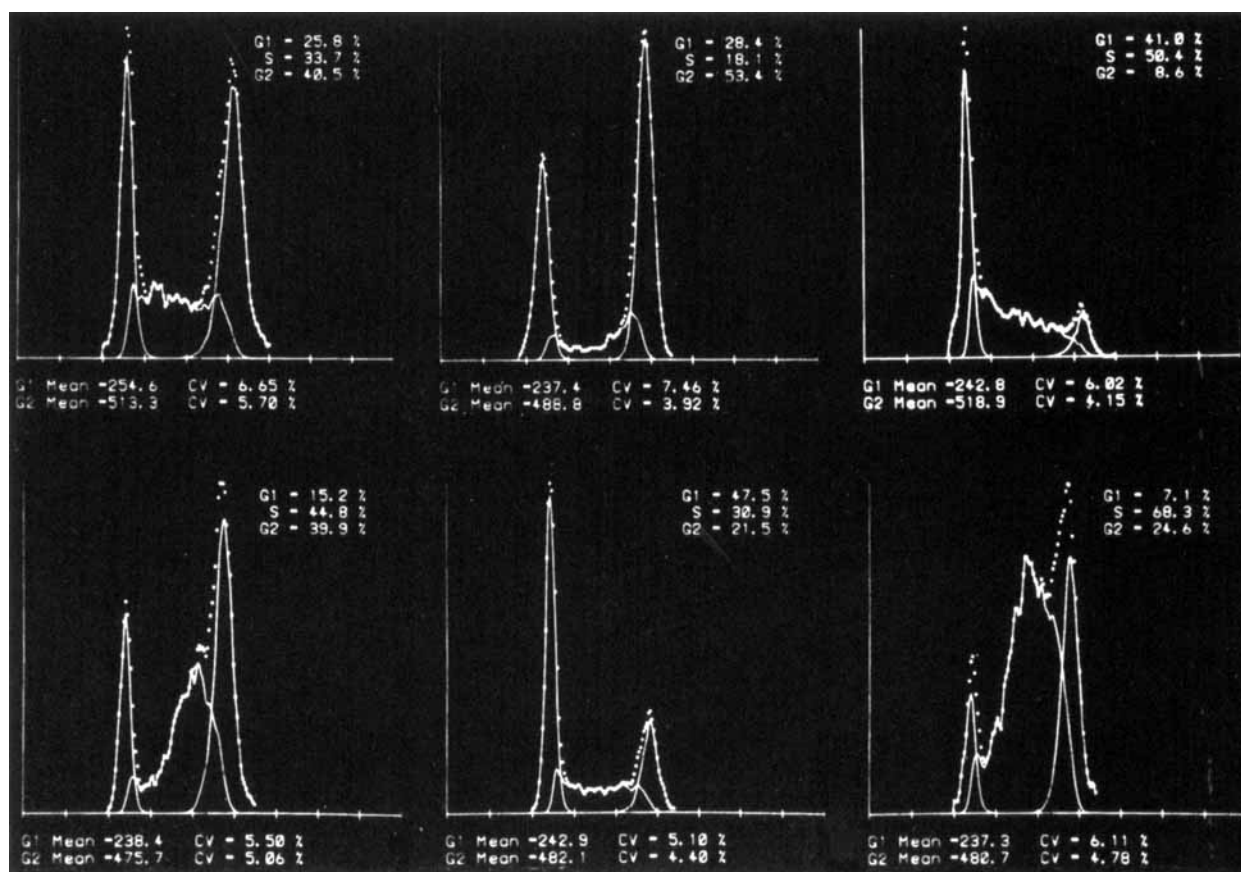


FIG. 5. A selection of histograms derived from experimental data. Display directly analogous to Figure 3.

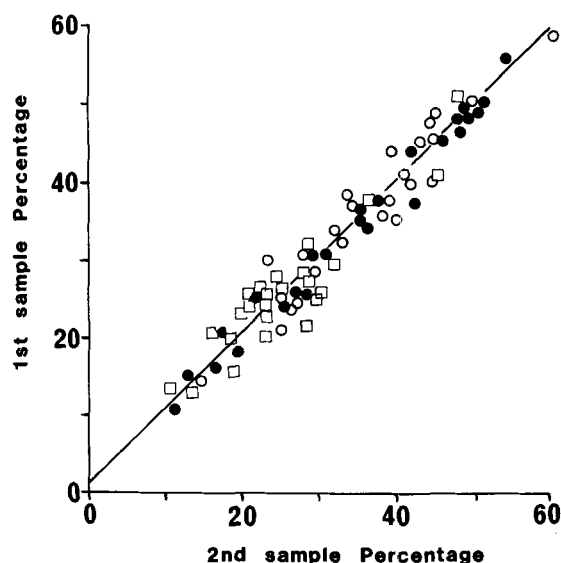


FIG. 6. Analysis of duplicate samples. Symbolic representation as for Figure 4. Data from only 25 analyses (75 points) shown, although the regression was calculated from 56 data sets (168 points). The slope and intercept were 0.97 and 1.04, respectively, with a correlation coefficient of 0.962.

of 75 points is shown in Figure 6, but the regression line was calculated from a total of 168 points, which gave a slope of 0.97 and intercept of 1.04 with 95% confidence limits of 5.2%. The correlation coefficient was 0.962. A $\Sigma\chi^2$ calculation for the deviation of one measurement from the other was 46 with 111 degrees of freedom, $p < 0.0001$ that the differences could have arisen by chance.

DISCUSSION

In the work presented in this paper, we have attempted to produce a method for analysing DNA histograms that is easy to use, robust, and fast on a microprocessor with only 28K addressable memory. As far as "user friendliness" is concerned, the program could hardly be easier. Only two numbers are required, the start and end channels over which the analysis is to take place. The procedure has been proven to be robust, with only two failures in over 350 analyses. In each of these two cases, there was no definable G1 peak, with the whole population being arrested in late S/G2+M after drug and radiation perturbation. In both cases, the program assumed that the first peak it defined was the G1 compartment when this was in fact the G2+M peak. It then attempted to find a G2+M peak in a region of the histogram where there were no data. This can be corrected by inserting a program queued request for

approximate positions for G1 and G2+M. The program has been found to execute rapidly, producing results within 20 to 30 s on a PDP LSI 11/23 microcomputer with only 28K addressable memory; the run times were dependent not only on the total number of channels over which the analysis had to be performed, but also on the spread in the data.

The analyses of the simulated data show that the method gives a good approximation for the proportions in the three phases (Fig. 4). The predicted proportions in G1 were almost identical with the true values over the whole range of from 15 to 60%. Also, the average position of the G1 mean calculated from the eight histograms analysed was predicted to within 0.2%. There tended to be a systematic error in the predicted proportions for S and G2+M with the former being underestimated and the latter being overestimated. This finding is consistent with the relatively larger error in the estimates of the G2+M coefficient of variation and the overall G2+M:G1 ratio, which was underestimated by 0.36%. This indicates that the model is not simulating the S/G2+M interface as well as it is simulating the G1/S interface in some types of histogram. The effect can be seen in the bottom left panel of Figure 3, where the G2+M:G1 ratio was markedly underestimated (1.948) and the G2+M CV was overestimated, 7.45% compared with 5.7%. The poor simulation of the S/G2+M interface in this histogram resulted in a negatively skewed G2+M distribution, which in turn gave rise to a shift of the mean to the left and the overestimate of the CV. These findings are consistent with those reported previously, where there also tended to be an overestimate of G2+M at the expense of S(3). However, the G2+M overestimate in this analysis was less than that in the overall results reported previously (3), and the overestimate of G1 at the expense of S has effectively been eliminated.

All analyses of DNA histograms are heavily dependent on accurate location of the G1/S and S/G2+M boundaries, and in this method the positions of the two interfaces are computed from equation 2 using the correction factors kG1 and kG2. The estimates of kG1 and kG2 are, in their turn, dependent on the ratios of the estimated frequencies of S-phase at the G1 and G2+M means and the frequencies of the G1 and G2+M populations at their respective means. These G1 and G2+M frequencies can be estimated with greater precision than can the S frequencies, as the latter are obtained by extrapolation from a region of the histogram at a 3-SD distance from the means. Thus, relatively small variations in the S distribution in the range over which the extrapolation is computed could make considerable differences to the predicted S frequency at the G1 and G2+M mean. Also, as the G2+M standard deviation will be approximately double that of G1 (constant CV), the potential errors in the estimate of the S frequency at the G2+M will be greater than those at the G1 mean. Generally, the G1 peak of a histogram is considerably higher than the G2+M peak, and there is not usually a correspondingly large difference in the frequency of the

S distribution at the G1 and G2+M means. Thus, any errors in the estimate of the S frequency at the G2+M mean will have a greater relative effect on the calculation of kG2 compared with the calculation of kG1 due to errors arising from the estimated S frequency at the G1 mean.

We have no direct evidence that the proportions of cells in the three phases from the experimentally derived data are correct. Indeed, direct evidence is almost impossible to obtain as there is no completely independent method of estimating the proportions in G1 and G2 directly, and in perturbed populations the S-phase fraction cannot be estimated reliably. In completely unperturbed populations, the ^3H -thymidine labelling index gives a good approximation for the proportions in S. But, in perturbed data sets (the majority of histograms analyses here), the labelling index must never be equated with the proportions in S, as a cell may have an S-phase DNA content and may not be synthesizing DNA due to the perturbing event. The majority of the experimental data sets exhibited a well-defined G1 peak, and the average G2+M:G1 ratio for over 350 analyses was 2.009, very close to the expected value. This is comparable to the result from the simulated data. The range of shapes of the experimental data was very similar to that found in the simulated histograms. We have no reason to suspect that any errors in the analysis of the experimental data were either qualitatively or quantitatively different from those in the simulated histograms and conclude that analysis of the former gives, at least, reasonable results. The comparison of duplicate samples would seem to support this claim. The result in Figure 6 indicates that the true proportion within a phase will have a 95% chance of being within $\pm 5.2\%$ of the computed value, and most investigators would accept a 5% biological variability between experiments. The duplicate samples analysed here contain more potential sources of variation than those due to staining, instrumental factors, and the analysis procedure. These samples were duplicated from the very start of each experiment; thus, additional sources of variation must be considered. These include dilution errors during seeding, growth variation from flask to flask, variations in radiation dose or drug concentration and exposure time (about 70% of these duplicates were perturbed populations), versene artefacts in single cell suspension preparation, dilution errors in preparing the suspension for staining, and possible modulation of the staining by the perturbing treatment under study. When all these factors are taken into consideration, it would seem unreasonable to expect reproducibility to be better than $\pm 5\%$ between samples. Reruns of the same samples gave results for the proportions within each phase to within $\pm 1.5\%$.

Baisch et al. recommended that 10^5 cells should be accumulated per histogram (3), but for these studies only between 5,000 and 10,000 cells were analysed. It would obviously be desirable to analyze more cells per data set, but many of our experiments involve 50 to 70 samples

at a time, and the data are collected list-mode, which consumes considerable quantities of disc space, and a compromise has to be drawn somewhere. High frequency noise can be a problem with relatively small numbers of cells analysed, but the high-frequency smoothing technique (21) has helped considerably. The close agreement between results from duplicate samples attests to the reliability of our methods in spite of the relatively small numbers of cells analysed per sample.

In comparison with many other techniques of analysing DNA histograms, this method is very simple, as it only requires the fundamental assumption that the data are normally distributed. It makes no assumptions concerning the age distribution of the population (22), which was one of the problems associated with our methods published previously (25,28), and no assumptions about the rate of DNA synthesis or of the G2+M:G1 ratio. The latter should obviously be 2.0; however, we only estimate DNA indirectly by flow cytometry. The quantity of fluorescent light emitted by a DNA-fluorochrome complex is dependent not only on the quantity of fluorochrome present but also on the number of accessible binding sites available per molecule of dye. The results that we see are governed by the law of mass action (20), and in order to think realistically about DNA histograms, we have to consider the number of dye molecules in relation to the number of accessible binding sites. Although the former can be controlled experimentally, the latter may not be controllable, as differences in chromatin structure and organization in G1, S, and G2+M cells may make differences to the quantity of fluorochrome that can be bound per DNA phosphate residue. In this series of experiments, the G2+M:G1 ratio was very close to the expected value of 2.0, suggesting that the staining procedure reflected stoichiometric binding irrespective of the perturbing events, that the instrument gave a linear response and that the method was capable of analysing and interpreting the data correctly. The linearity of response of the instrument has been checked by exploiting the coincidence phenomenon (24) as well as by using a pulse generator (unpublished). This was a particularly gratifying result, as absolutely no assumption about the G2+M:G1 ratio was made in the analysis process and the data included a considerable variation in the proportions of cells in the three phases.

The algorithms perform well for data sets with CVs of less than about 11%. As the SD of a distribution increases linearly with channel number in our instrument, a CV of 11% corresponds to the value that gives no clear distinction between the upper bound of the G1 peak and the lower bound of G2+M at the 3 standard deviations limit. Hence, the probability distribution for S-phase never reaches unity, as the G1 to G2+M interval (S-phase) corresponds to 9 G1 standard deviations, which is 3 G1 SDs + 3 G2+M SDs. A subsidiary routine has been written for this type of data, but this can only give suboptimal results. This condition has only arisen in 2% of samples, and typically the CVs were within the range of 3 to 7%.

Although there are some small systematic errors in this method of analysis, in practical terms these are likely to be less than those encountered because of biological variability not only within but also between experiments. However, the method cannot be used, as it stands at present, if a G1 peak cannot be defined; but we are currently investigating the possibility of producing a similar type of analysis for data sets of this type. The major advantage we have found with this method, apart from its simplicity and speed, is that it is capable of analysing drug- and radiation-perturbed data without any additional assumptions or conditions.

ADDENDUM

The algorithms described in this paper were written in Fortran IV to run on the DEC LSI 11/23 microcomputer. We have now upgraded one of our dedicated microprocessors with the 11/73 CPU and the programs execute in about 7–10 s with this processor. They have also been compiled with Fortran 77 to execute on a VAX 8600. A version has been written in Pascal in conjunction with Dr. M. Ormerod (Institute of Cancer Research, Sutton, UK) to run on the Ortho 2150 computer system. Similar versions are being prepared for both the Becton-Dickenson (FACS) and Coulter computer systems.

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